

the **maximum achievable prediction accuracy** $\Pi_{\max}$ is bounded by:

$$\Pi_{\max} \leq 1 - \frac{H_{\text{rate}}}{\log_2 N}$$

As entropy increases toward its maximum value $\log_2 N$, predictability approaches zero. Conversely, when entropy is near zero, forecasts can be nearly perfect.

Worked Examples

System	N	H_{rate} (bits/step)	$\Pi_{\max}$
Fair coin toss	2	1.0	$1 - \frac{1.0}{\log_2 2} = 0\%$
Loaded die*	6	1.5	$1 - \frac{1.5}{\log_2 6} \approx 41\%$
Periodic signal	100	0.01	$1 - \frac{0.01}{\log_2 100} \approx 99\%$

Table 2.3: Examples of entropy rate and resulting theoretical predictability. *Loaded die: non-uniform but not perfectly deterministic.

Key Implications

- **Perfect prediction** is only possible when $H_{\text{rate}} = 0$ (pure determinism). Example: Predicting sunrise using celestial mechanics.
- **Zero predictability** occurs when $H_{\text{rate}} = \log_2 N$ (maximal randomness). Example: Guessing lottery numbers.
- **Real-world examples:**
 - *Weather forecasting:* $\Pi_{\max} \approx 60\text{--}80\%$ — chaotic but partially deterministic.
 - *Stock prices:* $\Pi_{\max} \approx 10\text{--}30\%$ — high entropy due to external and stochastic factors.

Why This Matters

This inequality helps explain why:

- Weather forecasts degrade over time — chaos accumulates entropy.
- Financial markets resist prediction — high entropy reduces signal strength.
- Data compression is possible — low entropy signals are more predictable and redundant.

In summary, the entropy rate is not just a statistic—it defines the theoretical limit of what any forecasting model can achieve.

$$H_{\text{perm}} = - \sum_{j=1}^{m!} p_j \log_2 p_j, \quad H_{\text{norm}} = \frac{H_{\text{perm}}}{\log_2 m!}$$

PE evaluates the diversity of ordinal patterns in a time series. A lower PE suggests more deterministic dynamics.

In simple terms, PE looks at the relative order of consecutive values (patterns like “up-up-down”) and measures the diversity of such ordinal patterns. This method is especially appealing because it is invariant to monotonic transformations (e.g. scaling or shifting the data) – only the rank order matters – and it is computationally efficient even for large datasets.

Permutation Entropy: Definition and Interpretation

Calculation Procedure:

To compute the permutation entropy (PE) of a time series, follow these steps:

1. **Choose parameters:** Select an embedding dimension m (e.g., $m = 3$ or 4) and an embedding delay τ (typically $\tau = 1$).
2. **Extract ordinal patterns:** For each position i in the time series $x = \{x_1, x_2, \dots, x_N\}$ (from $i = 1$ to $N - (m - 1)\tau$), construct a window:

$$[x_i, x_{i+\tau}, x_{i+2\tau}, \dots, x_{i+(m-1)\tau}]$$

Rank the m values in ascending order to identify their relative order (i.e., ordinal pattern). For example, for $m = 3$:

- $x_i < x_{i+\tau} < x_{i+2\tau}$ corresponds to the pattern $(0, 1, 2)$
- $x_i < x_{i+2\tau} < x_{i+\tau}$ corresponds to the pattern $(0, 2, 1)$

There are $m!$ possible unique ordinal patterns of length m .

3. **Estimate probabilities:** Count how often each ordinal pattern occurs across the series to form a probability distribution:

$$\Pi = \{p_1, p_2, \dots, p_{m!}\}$$

4. **Compute Shannon entropy:**

$$H_{\text{perm}} = - \sum_{j=1}^{m!} p_j \log p_j$$

(Log base 2 is often used so the entropy is measured in bits.)

5. **Normalize (optional):** Divide by the maximum possible entropy $\log(m!)$ to obtain the normalized permutation entropy:

$$H_{\text{norm}} = \frac{H_{\text{perm}}}{\log m!}, \quad \text{so that } 0 \leq H_{\text{norm}} \leq 1$$

Interpretation of Extreme Cases:

- **Perfectly regular signal:** If the time series is completely deterministic (e.g., a constant or strictly increasing sequence), it produces only one ordinal pattern. In this case:

$$p_j = 1 \text{ for one pattern, and } p_k = 0 \text{ for all others} \Rightarrow H_{\text{perm}} = 0$$

This implies **low entropy**, indicating high regularity and predictability.